James Zhang

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Education

Yale University Sep 2025 – Dec 2025
Visiting Student, Economics, Graduate School of Arts and Sciences *GPA: 4.0/4.0*
PhD Coursework: Financial Economics, Econometrics

University of Maryland, College Park Aug 2022 – May 2025
B.S. Computer Science & Mathematics, Minor in Computational Finance *GPA: 3.9/4.0*
Coursework: Algorithms, Computer Networks, Deep Learning, Probability Theory, Mathematical Statistics, Linear Algebra

Publications

S. Nawathe, R. Panguluri, **J. Zhang**, S. Venkatesh. Multimodal Deep Reinforcement Learning for Portfolio Optimization. *arXiv preprint arXiv:2412.17293*, 2024.

Experience

Stripe Seattle, Washington
Software Engineer, Revenue Recognition *Nov 2026 – Present*

Brier Capital LLC Remote
Co-Founder, Quantitative Researcher *Apr 2026 – Present*

- Engineering a Rust trading system that combines market making and signal-based trading across Kalshi sports, crypto, and commodity markets, trading 200k+ contracts and earning $1,500 per week on $200/month of AWS infrastructure.
- Developing a sportsbook that answers live request-for-quote auctions for same-game parlays on Kalshi, handling 100M+ requests per day with bursts of 4,000 per second and pricing correlated legs across 10+ sports.
- Architecting the firm's AWS research platform, capturing market and alternative data into an S3 lake in Parquet queried through DuckDB on a cloud server, enabling large-scale backtesting before strategies trade live.

Yale University New Haven, Connecticut
Pre-Doctoral Research Fellow, Machine Learning and Financial Econometrics *Aug 2025 – Nov 2026*
Advisor: Professor Bryan Kelly, Head of Machine Learning at AQR

- Demonstrated that ridge regression forecasts on LLM-derived news signals earn 9.7% annual alpha (t=3.0) on 25 commodity markets, after transaction costs and controls for momentum, value, basis, and basis-momentum.
- Published Global Macroeconomic Data, a Python package that streams 1.5 TB of tick and macro data from LSEG Tick History, Datastream, WRDS, Compustat, OECD, and FRED through Polars into returns for 200+ global futures markets.
- Built an LLM pipeline that turns three decades of Dow Jones Newswires and Thomson Reuters articles into a dataset linking each article to its relevant futures markets and the paragraphs supporting each, running 30 million inference calls on Gemma 4 26B-A4B and Qwen 3.8-27B through vLLM.

Amazon Seattle, Washington
Software Development Engineering Intern *Jun 2025 – Aug 2025*

- Shipped video performance metrics, including 7-day click rates and average watch time, to the Video Shopping Experiences portal used by over 2 million sellers and vendors, built end to end from SQL to Java to a localized React frontend.
- Recommended migrating the portal's metrics store from Elasticsearch to AWS Timestream to meet its latency and cost targets, working with the Timestream team and presenting to principal engineers and management, who adopted it.

University of Maryland, College Park College Park, Maryland
Research Assistant, Machine Learning and Empirical Asset Pricing *Oct 2023 – May 2025*
Advisor: Professor Serhiy Kozak

- Modeled nonlinearities between firm characteristics with a kernel trick built in C++, CUDA, and Jax, achieving an annualized out-of-sample Sharpe ratio of 3.2 against 1.65 for PCA on 40 anomaly portfolios.
- Designed a Python package for PARAFAC tensor decompositions with Orthogonalized Alternating Least Squares, running 20% faster than TensorLy using Jax on a GPU.

Capital One McLean, Virginia
Software Engineering Intern *Jun 2024 – Aug 2024*

- Deployed a FastAPI Slackbot to production on AWS ECS Fargate that replaced emailed requests to the Data Remediation team with a tracked Slack workflow, so requesters got answers sooner and engineers stopped losing time to email.

Skills

Languages: Python, Rust, Java, C++, CUDA, SQL
Libraries: PyTorch, Scikit-learn, Statsmodels, Jax, vLLM, Transformers, Pandas, NumPy, Polars
Tools: AWS (S3, Lambda, EC2), SLURM, Hugging Face, Elasticsearch, Kalshi API, WebSockets, Claude Code